Sync with Eden: draft results

28 October 2026 · TUM Quantum Hackathon, Finance track · Allocation route (3 of 6 industries) · Final pitch 30 October

> Growth switches, risk-sensitive keeps, and balanced (the default profile) goes to review, because its answer flips inside the estimation error. Every number below is real data from one reproducible run (`python scripts/run_all.py run --canonical`, about 22 minutes, fixed seeds); no synthetic data was needed.

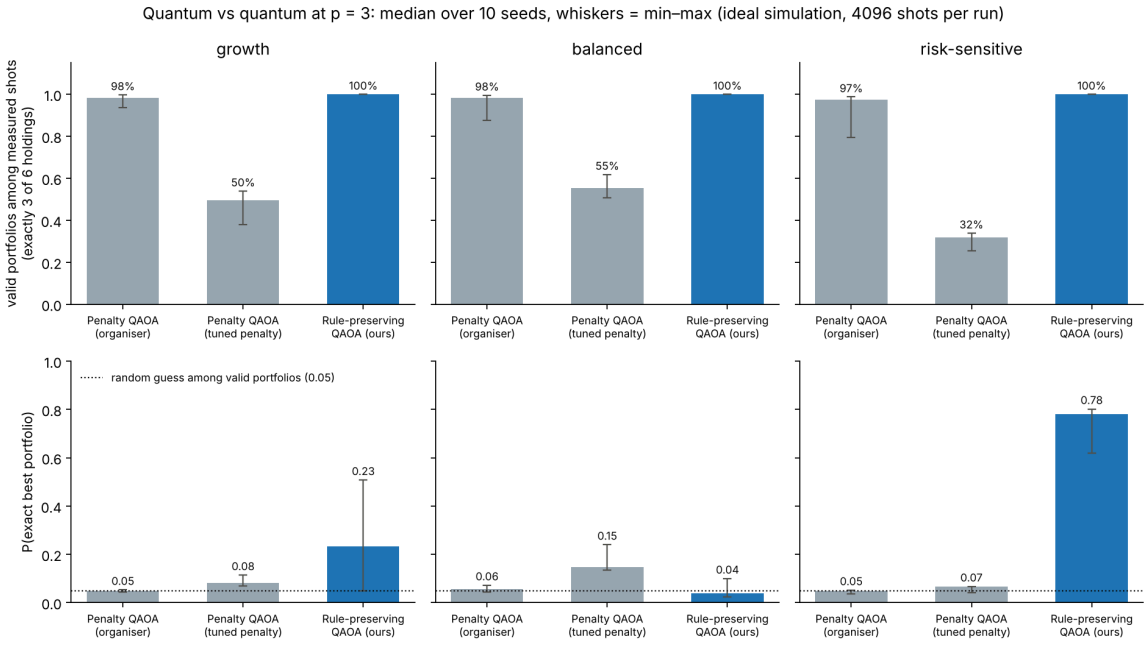
At a glance

The fund's satellite tilt holds NoDur + Hlth + Utils; the proposal swaps NoDur for HiTec. Switching needs the same winner in all 3 scenario sets and at least 70 % support in 500 bootstrap resamples.

Profile	Verdict	Evidence
Growth	**Switch** to HiTec + Hlth + Utils	Same winner in all 3 scenario sets; beats current in 89 % of resamples
Balanced (default)	**Review**, hold for now	Winner flips under the supplied return stress; proposal better in 74 % of resamples
Risk-sensitive	**Keep** NoDur + Hlth + Utils	Winner in all 3 sets; passes the loss rule in 76 % of resamples

Plot 1: success rate, quantum vs quantum

Our rule-preserving QAOA puts 100 % of measured shots on valid 3-of-6 portfolios by construction, and finds the best portfolio far more often where the decision is clear (risk-sensitive 0.78 vs 0.05). Ideal simulation, 3 layers, 10 seeds, 4096 shots per run; random guessing among the 20 valid portfolios gives 0.05.



results/canonical/figures/fig_success_rate.png

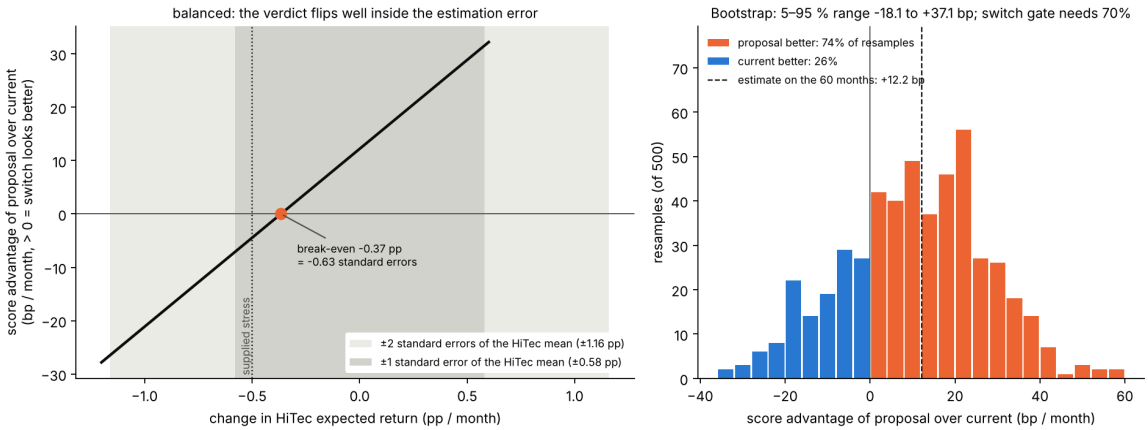
Profile	Valid shots: ours / organiser / tuned	P(best), median: ours / organiser / tuned
Growth	100 % / 98 % / 50 %	0.23 / 0.05 / 0.08
Balanced	100 % / 98 % / 55 %	0.04 / 0.06 / 0.15
Risk-sensitive	100 % / 97 % / 32 %	**0.78** / 0.05 / 0.07

- Penalty trade-off: the organiser's large penalty keeps shots valid but leaves P(best) at guessing level; the tuned penalty recovers the signal but wastes about half the shots. Ours avoids the trade-off.
- Growth is better but unsteady across seeds (0.05 to 0.51).
- Balanced is a near-tie and ours is not better there: 0.04, below guessing in 6 of 10 seeds. The CVaR variant reaches 0.25 and decodes the right answer in all 10 runs.
- With shot noise during optimisation, ours drops to 0.36 (risk-sensitive) and 0.02 (growth).

Proposed slide line: the rules are part of the circuit, so the search can never propose an illegal portfolio; where the decision is a near-tie, every method struggles.

Plot 2: balanced break-even vs estimation error

For the balanced profile the switch decision flips at 0.63 standard errors of the HiTec estimate, so the data cannot tell the two portfolios apart.



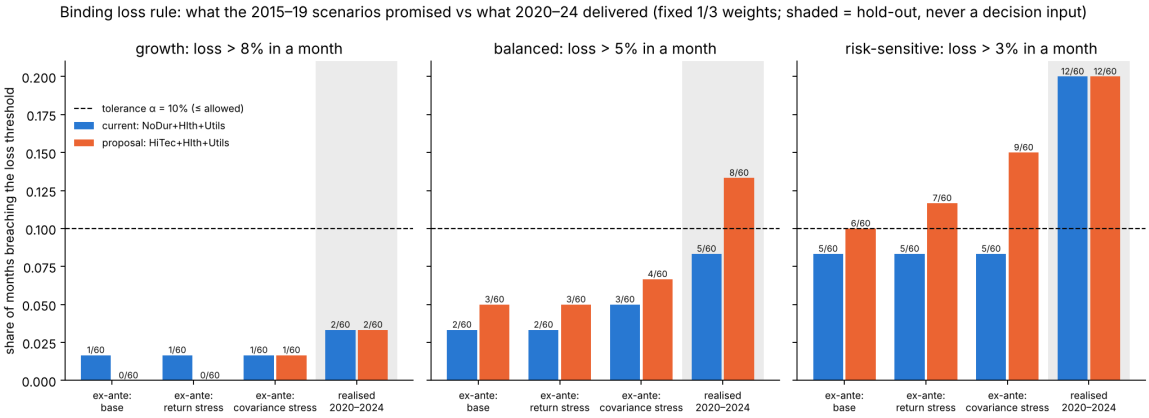
results/canonical/figures/fig_estimation_error.png

- On the 2015-19 estimates the proposal scores 12 bp per month better than the current tilt.
- The lead disappears if HiTec's expected return is 0.37 points per month lower. The estimate (1.51 % per month) has a standard error of 0.58 points.
- The supplied return stress (−0.5 points, 0.86 standard errors) lies beyond the break-even, so the winner flips: review.
- Bootstrap, 500 resamples: proposal better in 74 %; 5-95 % range of its lead −18 to +37 bp per month.
- Other balanced break-evens: switching cost 0.0057 (base 0.002); risk weight 7.5 and 14.9 (base 5).
- Growth flips only at −0.60 points (1.04 standard errors, beyond the stress); risk-sensitive would need HiTec 0.55 points better.

Proposed slide line: for the default profile the honest answer is review, plus the one belief the committee must state: does tech keep its edge?

Plot 3: loss-limit check, 2020-24

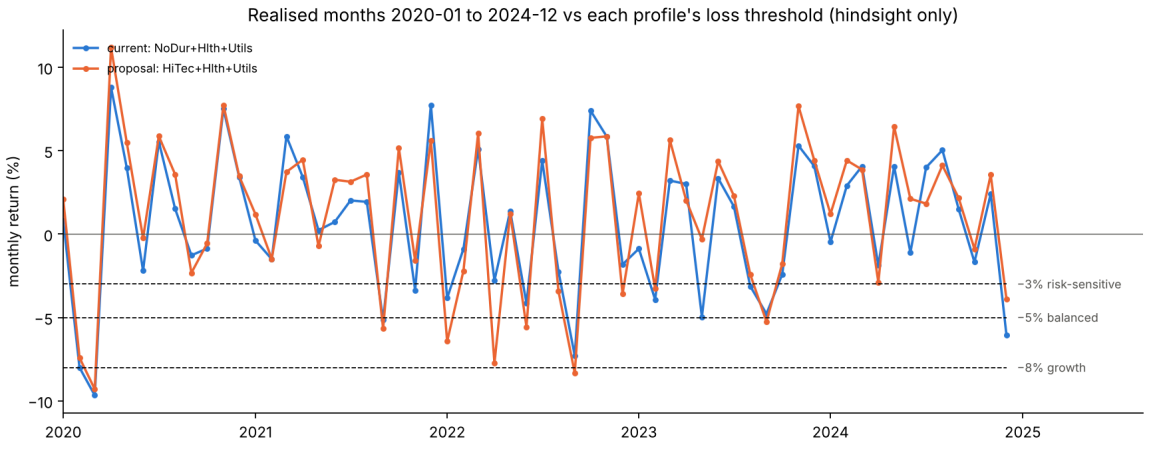
The loss rule held for growth but not for the cautious profile: risk-sensitive breached its 3 % limit in 12 of 60 months (20 %), twice the 10 % tolerance, although every ex-ante scenario set said it passed. Fixed one-third weights (the rule's own definition); 2020-24 never fed a decision.



results/canonical/figures/fig_loss_limits.png

Profile (threshold)	Portfolio	2015-19 base	Return stress	Covariance stress	Realised 2020-24
Growth (8 %)	current	1	1	1	2 (pass)
Growth (8 %)	proposal	0	0	1	2 (pass)
Balanced (5 %)	current	2	2	3	5 (pass)
Balanced (5 %)	proposal	3	3	4	**8 (fail)**
Risk-sensitive (3 %)	current	5	5	5	**12 (fail)**
Risk-sensitive (3 %)	proposal	6	7 (fail)	9 (fail)	**12 (fail)**

Breach months out of 60; at most 6 allowed.



results/canonical/figures/fig_loss_timeline.png

- The 60-month scenario table understated tail risk (COVID crash in 2020, 2022 drawdown).
- For balanced the proposal would have broken the rule and the current tilt would not: the review verdict was the safer call.
- Hindsight wealth after switching cost: proposal 1.86 vs current 1.38, drawdown −21 % vs −17 %. One path only; it neither proves nor refutes the 2019 decision.

Situation changes

One changed assumption re-runs in about a minute (`--fast --set ...`) and `diff` names what flipped and why.

Change	What the diff shows
Switching cost ×3 (`change_penalty=0.006`)	Balanced: review to keep. Growth: switch to review (flips under the return stress)
Balanced loss threshold 4 % (`balanced.loss_threshold=0.04`)	No verdict changes; balanced admissible set shrinks from 19 to 15 portfolios

Synthetic data: our position

We do not need synthetic data: all three plots run on real data in one reproducible command, so the deck is 100 % real, 0 % synthetic.

- Synthetic numbers on a decision slide would cost us on the graded criteria (traceability, reproducible GitHub code, honest limits) and contradict our responsibility story: a validation team must be able to reproduce every number.
- The only defensible use would be a clearly labelled synthetic scaling instance (10 to 20 random assets) to show how the circuits grow, never feeding a verdict.
- Bootstrap resamples are not synthetic: they are the real 60 months redrawn, and we call them that.
- Labels we do need: "ideal simulation" vs "with shot noise", and "decision data 2015-19" vs "hindsight 2020-24".
- Re-run on a second machine: exact and bootstrap results bit-identical; QAOA medians within about 0.01.

Questions for Eden

☐ Story order: lead with the decision (keep / switch / review per profile) and show quantum as a verified tool, or lead with the quantum comparison? Our plan: decision first.

☐ Is "review" convincing as the headline for the default (balanced) profile, or does the jury expect one answer?

☐ Quantum claim strength: "100 % valid by construction, far better where the decision is clear, no better on a near-tie". Right level?

☐ Hindsight loss-rule failure: present it as a model-risk finding in the recommendation, or keep it on the limits slide?

☐ Synthetic data: we propose none. Is there any slide where a labelled synthetic example would help?

☐ Policy Practice Team feedback: we document two changes (profiles chosen actively, not by age; tech cap as a preference). Enough, or show one live in the model?

☐ Individual assessment: how to split the 20 minutes so each member explains their own part?

Limits to say out loud

60 months of US industry data; equal weights and exactly 3 holdings; the switching cost is a proxy, not a fee model; mandate numbers are organiser assumptions; ideal simulation on a laptop. Twenty options are solved exactly in milliseconds, so we claim no quantum advantage.

Source of every number: `results/canonical/` on branch `claude/draft-plots-eden-sync` of the team repo.